GFP I

Protocol for the Biochemistry lab course by
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1 Introduction

(Rathert and Jeltsch, 2025).

2 Materials and methods

All experiments were performed as described in the course manual unless stated otherwise (Lab Manual, 2025). The deviations from the course manual are described below. The centrifugation of the crude extract was already performed by the supervisor, so the remaining supernatant was directly applied to the column.
The flow through fraction was collected right after the application of the GFP sample directly to the Eppendorf tube instead of collecting the whole flow through at first and then taking 10 µL of it for the SDS-PAGE analysis.
Additionally, the volumes of the components of the resolving and stacking gel were altered according to the instructions of the supervisor, which are listed in Table 1.

Tab. 1: Components for the gels of one prepared SDS-PAGE.

resolving gel		stacking gel	
reagent	volume	reagent	volume
1.5 M Tris-HCl, pH 8.8	2 mL	0.5 M Tris-HCl, pH 6.8	1 mL
ddH_2O	1.8 mL	ddH_2O	2.24 mL
10% SDS	80 µL	10% SDS	40 µL
30% AA/BA	4 mL	30% AA/BA	0.68 mL
40% APS	16 µL	40% APS	8 µL
TEMED	16 µL	TEMED	8 µL

Furthermore, the molecular weight marker was loaded into well 2 of the gel stained with Coomassie Blue, while it was loaded into well 1 of the gel for Western Blot analysis.

3 Results

In this experiment, the GFP with a His-tag was purified using Ni-NTA affinity chromatography and analyzed by SDS-PAGE and Western Blot. Additionally, the absorption spectra of the eluted fractions were recorded to determine the concentration of the purified GFP in the fractions.

3.1 UV/Vis spectroscopy

One method to determine the concentration of a protein in a solution is to measure the absorption at 280 nm and to calculate the concentration using the Beer-Lambert law described in Equation 1.

$$A_\lambda = \varepsilon_\lambda \cdot c \cdot l \tag{1}$$

where A_λ is the absorption at wavelength λ, ε_λ is the molar extinction coefficient, c is the concentration and l is the path length of the cuvette, which is usually 1 cm. By rearranging the equation and inserting $\varepsilon_{280\,\mathrm{nm}} = 22\,000\,\frac{\mathrm{L}}{\mathrm{mol \cdot cm}}$ as the molar extinction coefficient for GFP (Lab Manual, 2025), the concentration of each fraction can be calculated as shown in Equation 2.

$$c = \frac{A_{280\,\mathrm{nm}}}{22\,000\,\frac{\mathrm{L}}{\mathrm{mol \cdot cm}} \cdot 1\,\mathrm{cm}} \tag{2}$$

The results of the UV/Vis spectroscopy with NanoDrop are shown in Table 2 with the absorption at 280 nm, the calculated concentration of GFP and the quotient of the absorption at 260 nm and 280 nm.

Tab. 2: Results of the UV/Vis spectroscopy with NanoDrop and the caluculated concentrations of GFP in elution fractions E1-E6.

fraction	$A_{280\,\mathrm{nm}}$	c / $\frac{\mu\mathrm{mol}}{\mathrm{L}}$	$\frac{A_{260\,\mathrm{nm}}}{A_{280\,\mathrm{nm}}}$
E1			
E2			
E3			
E4			
E5			
E6			

The absorption and concentration values from Table 2 show that the concentration of GFP in the elution fraction E2 is the highest and that it decreases in the following fractions until no absorption is detected in E5 and E6, indicating that the purified GFP is mostly present in E2 and E3 and to a smaller extent in E4. The quotient of the absorption at 260 nm and 280 nm is an indicator for the purity of the protein solution, where a value of around 0.6 indicates a pure protein solution and higher values indicate

contamination with DNA (DeNovix, 2025). The values of the quotient in Table 2 show that the elution fractions E2 and E3 have a high purity with quotients of 0.56 and 0.63, while the other fractions have higher values, indicating contamination.

3.2 Molecular weight determination

The stained gel in **??** shows one smeared band in the lane of the flow through, which can be explained by the altered protocol of collecting only a small fraction of the flow through and therefore not loading all proteins from the GFP supernatant on the gel.

4 Discussion

5 References

Books and Articles

Lab Manual (2025). *Experiment GFP I*. Universität Stuttgart. Stuttgart.

Rathert, Philipp and Albert Jeltsch (2025). *Experiment GFP I*. Universität Stuttgart. Stuttgart.

Internet Sources

DeNovix (2025). *Purity Ratios Explained*. URL: https://www.denovix.com/tn-130-purity-ratios-explained/#:~:text= (visited on 14/03/2026).